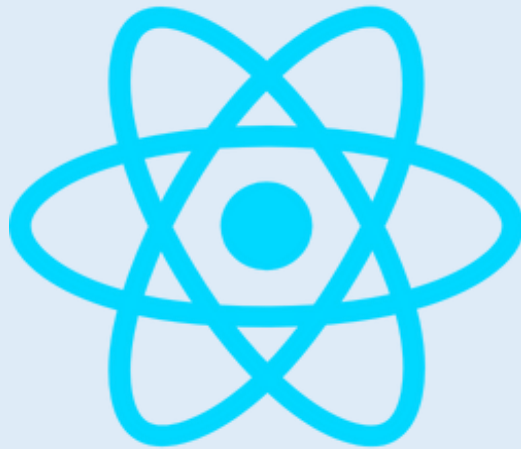
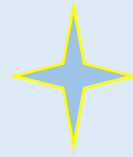
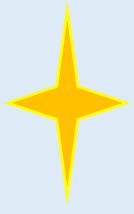


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#5 – ReactJs





ReactJs is a Front-end Javascript Library

React is a Javascript library For building user interfaces

React allows us to create reusable UI components

React was developed by the facebook software engineer Jordan Walke



Javascript

Small Project

Structure Free prone to mess

Changes Element manually

Reactjs

Large Project

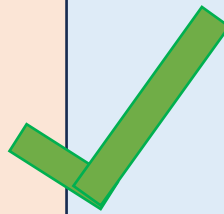
Structure by Component leads to neat

Changes Element based On State



with-react

your-project
node_modules
public
src
components
 Card.jsx
 Button.jsx
 Input.jsx
features
 List
 List-create.jsx
 List-update.jsx
 List-read.jsx
 List-delete.jsx
 User
 Login.jsx
 Register.jsx
main.jsx
package.json



without-react

your-project
Index.html
style.css
app.js

Packages ??
Structure ???



React JSX

JSX stands for Javascript XML

JSX allows us to write HTML in React

JSX works perfect inside react components

With `{}` we can do operations

React Components are like function that return HTML elements

We can create react components with ES7 +



Example React Components :

```
const paragraphs = ( props ) => {  
  return (  
    <div>  
      <p> The total is {props.text} </p>  
    </div>  
  )  
}  
export default paragraphs
```



React Event

Just Like HTML DOM events, React can perform actions based on user events

React has same events as HTML : click, change, mouseover, drag, etc

Example React event with component :

```
const button = () => {  
  const shoot = () => {  
    alert("Shoot")  
  }  
  return (  
    <div onClick={shoot}>  
      <button>Shoot </button>  
    </div>  
  )  
};  
export default button
```



React Hook

React Hook is A feature in React that allows us to do something in component

useState – storing data

useEffect – side effects

useContext – share data globally

useRef – accessing dom



useState – storing data

Syntax :

```
import { useState } from "react"
const [ state , setState ]=useState(initialState)
```

Example :

```
import { useState } from "react"
const counter = () => {
  const [ count, setCount ] = useState(0)
  return (
    <div>
      <p> Count : {count } </p>
      <button onClick={()=>setCount(count+ 1)} >
        Add
      </button>
    </div>
  )
};
```



useEffect – side effects

Syntax :

```
useEffect(<function/>,[dependency]);
```

1. No passed dependency

```
useEffect(()=>{ console.log("runs on every render") })
```

2. An empty array

```
useEffect(()=>{ console.log("runs only the first render")},[])
```

3. Props or state values

```
useEffect(()=>{ console.log("runs only the first  
render and  
any time the dependency changes")},[value])
```



useEffect – example

```
import { useEffect } from "react"

const Card = () => {
  useEffect(() => {
    console.log("runs only the first render"); }
    , [])
  return (
    <div>
      This is Component Card
    </div>
  )
};

export default Card
```



```
const ComponentA = () => {  
  const [currentUser, setCurrentUser] = useState({name:"josse"});  
  return (  
    <div>  
      <ComponentB currentUser={currentUser} />  
    </div>  
  )  
};
```

```
const ComponentB = ( {currentUser} ) => {  
  return (  
    <div>  
      <ComponentC currentUser={currentUser} />  
    </div>  
  )  
};
```

```
const ComponentC = ( {currentUser} ) => {  
  return (  
    <div>  
      {currentUser}  
    </div>  
  )  
};
```



useContext – share data globally

```
import { createContext , useContext , useState } from "react"
```

```
const AuthContext = createContext(null)
```

```
const App = () => {
```

```
  const [ currentUser , setCurrentUser ] = useState({name: "josse"})
```

```
  return (
```

```
    <AuthContext value={currentUser} >
```

```
      <Component />
```

```
    </AuthContext>
```

```
  )
```

```
};
```

```
const component = () => {
```

```
  const login = useContext(AuthContext)
```

```
  return (
```

```
    <div>
```

```
      Welcome { login.name }
```

```
    </div>
```

```
  )
```

```
};
```



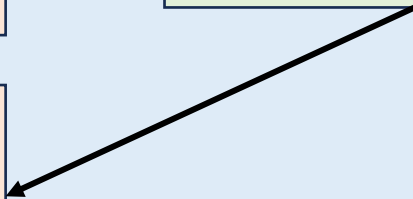
with useContext()

A

B

C

useContext()



without useContext()

A

B

C





useRef – accessing DOM

```
import { useRef } from "react"

const App = () => {
  const inputRef = useRef(null)
  return (
    <div>
      <button onClick={() => inputRef.current.focus()}>
        Click Me !
      </button>
      <input ref={inputRef} />
    </div>
  )
}
```



THANK YOU !

Don't forget to visit

<https://josserich.github.io>